

CEDLEARN

Crop Disease Detection Using Machine Learning

Capstone Project — Problem Statement

Domain Agriculture & AgriTech	Level Intermediate	Type Multi-Class Classification	Dataset 5,000 rows
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1. Background

Agriculture is the backbone of India's economy. One of the most critical threats to crop productivity is disease — caused by fungal pathogens, bacterial infections, and pest infestations. Early identification of disease type can reduce crop loss and guide timely farmer intervention.

In this project, you will build a machine learning model that classifies a crop observation into one of four categories based on environmental sensor data, soil measurements, and farm management records.

2. Problem Statement

You are a junior data scientist at an AgriTech startup. The company has deployed IoT sensors across partner farms. Your task is to build a multi-class classification model to predict the type of crop disease (or confirm the crop is healthy) using the provided dataset.

Target Variable: `disease_label`

Class	Description	Action Required
Healthy	No disease present	Routine monitoring
Fungal	Fungal pathogen (blight, rust)	Apply fungicide
Bacterial	Bacterial infection (wilt, leaf spot)	Bactericide + quarantine
Pest Damage	Insect or pest damage	Pesticide + bio-control

3. Dataset Description

File: `crop_disease_dataset.csv` | Rows: 5,000 | Columns: 18 (17 features + 1 target)

Note: The dataset is intentionally messy. It contains missing values, inconsistent capitalisation in categorical columns, outlier values in numerical columns, and physically implausible entries. Handling all data quality issues is part of this project.

Feature Name	Type	Description
<code>crop_type</code>	Categorical	Type of crop (Rice, Wheat, Maize, Cotton, Soybean)
<code>region</code>	Categorical	Geographic region of the farm
<code>season</code>	Categorical	Growing season (Kharif / Rabi / Zaid)
<code>soil_type</code>	Categorical	Soil type (Loamy, Sandy, Clay, Silty)
<code>irrigation_method</code>	Categorical	Irrigation type (Drip, Flood, Sprinkler, Rainfed)

temperature_C	Numerical	Average daily temperature in Celsius
humidity_percent	Numerical	Relative humidity (%)
rainfall_mm	Numerical	Total rainfall in mm over the past week
leaf_wetness_hrs	Numerical	Hours of leaf surface wetness per day
soil_pH	Numerical	Soil pH level (3–10 is valid range)
nitrogen_kg_ha	Numerical	Nitrogen content in soil (kg/hectare)
phosphorus_kg_ha	Numerical	Phosphorus content in soil (kg/hectare)
potassium_kg_ha	Numerical	Potassium content in soil (kg/hectare)
days_since_last_rain	Numerical	Days elapsed since last recorded rainfall
plant_age_days	Numerical	Age of the plant in days from sowing
pesticide_applied	Binary	Whether pesticide was applied (1 = Yes, 0 = No)
fertilizer_kg_ha	Numerical	Total fertilizer applied (kg/hectare)
disease_label	Target	Healthy / Fungal / Bacterial / Pest Damage

4. Tasks

Complete all tasks below in a Jupyter Notebook. Each task must include code, output, and a brief markdown explanation of your observations.

Type	Task
EDA	Load the dataset. Report shape, data types, and missing values per column.
EDA	Check all categorical columns for inconsistent capitalisation (e.g. 'WHEAT' vs 'wheat'). List all unique values found.
EDA	Plot the class distribution of disease_label. Is the dataset balanced? Discuss what imbalance means for a classifier.
EDA	Plot histograms for temperature_C and humidity_percent. Identify physically implausible outlier values.
EDA	Plot a boxplot of soil_pH grouped by disease class. What differences do you observe?
EDA	Compute and plot a correlation heatmap for all numerical features. Identify the most correlated pair.
Clean	Standardise all categorical columns to Title Case. Verify with unique value counts after cleaning.
Clean	Replace out-of-range values with NaN using domain knowledge (e.g. temperature must be 0–55°C).
Clean	Impute missing values: median for skewed columns, mean for normal columns, mode for categorical columns.
FE	Create HTI = (humidity_percent * temperature_C) / 100 to capture combined heat-humidity stress.
FE	Create drought_stress = 1 if days_since_last_rain > 20 AND rainfall_mm < 10, else 0.
FE	Create maturity_stage column: Seedling (<30 days), Vegetative (30–70), Reproductive (>70). Label encode it.
FE	One-Hot Encode all remaining categorical columns. Explain why OHE is preferred over Label Encoding here.
Model	Split data 80/20 (train/test). Build a Decision Tree, KNN (try k=3,5,7), and Random Forest classifier.
Model	Print the Classification Report for each model. Which class has the lowest F1-score and why?
Model	Plot a Confusion Matrix for the best model. Which two classes are most commonly confused?

Model

Create a comparison table: Model | Train Acc | Test Acc | Macro F1. Recommend one model for deployment with justification.

5. Deliverables

1. Cleaned and feature-engineered dataset saved as a .csv file
2. Fully executed Jupyter Notebook (.ipynb) with all code, outputs, and markdown explanations
3. At least 6 visualisation plots (EDA + model evaluation)
4. Final model comparison table with a written recommendation paragraph

